# How Crossing Over Creates New Allele Combinations

Biology 2e chapter 11 reading connection | Teacher guide / classroom preparation

## Learning target

How does exchanging chromosome segments change the allele combinations in gametes?

## Timing and materials

One class period; pairs or groups of three. Print and cut strips beforehand to save time.

Original printable chromosome strips, two colors of paper or colored pencils, scissors, tape, and four paper circles for gametes. Reusable segmented beads are an alternative.

## Classroom route

1. Label two identical strips A-B-C and two a-b-c. Use one color per homolog, and mark a centromere to the left of A. Pair sister strips at their centromeres, then align the two homologs gene for gene.

2. First model division without an exchange. Repeat with one exchange between A and B, then between B and C. Cut and reciprocally exchange equal end segments of one chromatid from each homolog. Leave the other two chromatids unchanged.

3. Separate homologs in meiosis I, then sister chromatids in meiosis II. Place one chromatid in each of four gametes. Record the four allele strings and tally parental versus recombinant combinations relative to ABC and abc. Compare the three models.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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## Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

## Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

## Expected reasoning

The no-exchange model yields ABC, ABC, abc and abc. An exchange between A and B yields ABC, Abc, aBC and abc; an exchange between B and C yields ABC, ABc, abC and abc (order varies with chosen chromatids). Each single exchange produces two parental and two recombinant chromatids. Recombination rearranges existing alleles; it does not create new ones. This controlled model does not estimate crossover frequency in organisms.

## Feedback and scoring

| Criterion | Points | Evidence |
|---|---|---|
| Prediction and controls | 0-2 | Testable claim; comparable samples and controlled variables. |
| Observations | 0-2 | Complete, labeled records with units and source. |
| Interpretation | 0-2 | Correct calculations or categorical comparisons; claim tied to evidence. |
| Limitations | 0-2 | Specific uncertainty and a useful follow-up investigation. |

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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## Model limitations

A manipulable chromosome model, not observed cell data. One homologous pair, three loci, and at most one crossover are represented. The no-exchange comparison isolates the effect of recombination; real meiosis includes additional chromosome pairs, assortment and crossover constraints. The model does not predict real recombination frequencies.

## Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

## Evidence and optional classroom resources

Classroom chromosome modeling reference

https://teach.genetics.utah.edu/content/genetics/inheritance/

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

## Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.